describe a method to consistently estimate the latent classes k_{it} from the data, under the assumption that firm heterogeneity has a finite, known number of points of support in the population.

Workers are also heterogeneous, and we denote the *type* of worker i as α_i. These types can be discrete or continuous, depending on the specification. In addition to their unobserved types, workers may also differ in terms of their observable characteristics X_{it}.

Last, worker i receives log-earnings Y_{it} at time t. The observed data for worker i are thus a sequence of earnings $(Y_{i1}, \ldots, Y_{iT})$, firm and mobility indicators $(j_{i1}, m_{i1}, \ldots, j_{i,T-1}, m_{i,T-1}, j_{iT})$, and covariates $(X_{i1}, \ldots, X_{iT})$. We consider a balanced panel setup for simplicity, and we focus on workers receiving positive earnings in each period.

In this framework, we will be interested in recovering the distributions of log-earnings for workers of type α in firms of class k, and the proportions of type-α workers in class-k firms. Earnings distributions will be informative about complementarities, while type proportions will be informative about sorting patterns. In addition, within our framework we will be able to document transition probabilities and other dynamic aspects.

We consider two different models: a static model where current earnings do not affect job mobility or future earnings conditional on worker type and firm class, and a dynamic model that allows for these possibilities. We now describe these two models in turn. Next, we discuss how our assumptions map to theoretical sorting models proposed in the literature. Throughout, we denote $Z_i^t = (Z_{i1}, \ldots, Z_{it})$ the history of a random variable Z_{it} up to period t.

2.1. *Static Model*

There are two main assumptions in the static model. First, job mobility may depend on the type of the worker and the classes of the firms, but not directly on earnings. As a result, the firm and mobility indicators, and firm classes, are all *strictly exogenous* in the panel data sense. In addition, covariates are also strictly exogenous. Second, log-earnings after a job move are not allowed to depend on previous firm classes or previous earnings, conditional on the worker type and the new firm's class.

Before stating the assumptions formally, let us describe the model's timing. In period 1, the type of a worker i, α_i, is drawn from a distribution that depends on the class k_{i1} of the firm where she is employed and her characteristics X_{i1}. The worker draws log-earnings Y_{i1} from a distribution that depends on α_i, k_{i1}, and X_{i1}.

At the end of every period $t \geq 1$, the worker moves to another firm (i.e., $m_{it} = 1$ or 0) with a probability that may depend on her type α_i, her characteristics X_i^t, the fact that she moved in previous periods m_i^{t-1}, and current and past firm classes k_i^t. This probability, like all other probability distributions in the model, may depend on t unrestrictedly. Moreover, the probability that the class of the firm she moves to is $k_{i,t+1} = k'$ may also depend on α_i, X_i^t, m_i^{t-1}, and k_i^t (while also varying with k'). Last, covariates $X_{i,t+1}$ are drawn from a distribution depending on α_i, X_i^t, m_i^t, and k_i^{t+1}.

If the worker changes firm (i.e., when $m_{it} = 1$), log-earnings $Y_{i,t+1}$ in period $t+1$ are drawn from a distribution that depends on α_i, $X_{i,t+1}$, and $k_{i,t+1}$. If instead the worker remains in the same firm between t and $t+1$ (i.e., $m_{it} = 0$), $Y_{i,t+1}$ are drawn from an unrestricted distribution that may depend on Y_i^t, α_i, X_i^{t+1}, and k_i^{t+1}.

Formally, the two main assumptions are thus as follows.

ASSUMPTION 1—Static Model:

(i) (*Mobility Determinants*) m_{it}, $k_{i,t+1}$, *and* $X_{i,t+1}$ *are independent of* Y_i^t *conditional on* α_i, k_i^t, m_i^{t-1}, *and* X_i^t.

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(ii) (*Serial Independence*) $Y_{i,t+1}$ *is independent of* Y_i^t, k_i^t, m_i^{t-1}, *and* X_i^t *conditional on* α_i, $k_{i,t+1}$, $X_{i,t+1}$, *and* $m_{it} = 1$.

A simple example of the static model is the following log-earnings regression:

$$Y_{it} = a_t(k_{it}) + b_t(k_{it})\alpha_i + X_{it}'c_t + \varepsilon_{it}, \tag{1}$$

where $\mathbb{E}(\varepsilon_{it}|\alpha_i, k_i^T, m_i^T, X_i^T) = 0$. This model simplifies to the AKM model in the absence of interaction effects, that is, when $b_t(k) = 1$, and firms j_{it} and classes k_{it} coincide.[5]

2.2. *Dynamic Model*

There are two main differences between the dynamic model and the static model. First, at the end of period t, the worker moves to another firm with a probability that depends on her current log-earnings Y_{it}, in addition to her type α_i, X_{it}, and k_{it}, and likewise the probability to move to a firm of class $k_{i,t+1} = k'$ also depends on Y_{it}. Second, log-earnings $Y_{i,t+1}$ in period $t+1$ are drawn from a distribution depending on the previous log-earnings Y_{it} and the previous firm class k_{it}, in addition to α_i, $X_{i,t+1}$, and $k_{i,t+1}$. Job movers and job stayers draw their log-earnings from different distributions conditional on these variables. As we discuss in the next subsection, allowing for these features is important in order to nest a number of structural models of wage and employment dynamics that have been proposed in the literature. Formally, we make the following assumptions.

ASSUMPTION 2—Dynamic Model:

(i) (*Mobility Determinants*) m_{it}, $k_{i,t+1}$, *and* $X_{i,t+1}$ *are independent of* Y_i^{t-1}, k_i^{t-1}, m_i^{t-1}, *and* X_i^{t-1} *conditional on* Y_{it}, α_i, k_{it}, *and* X_{it}.

(ii) (*Serial Dependence*) $Y_{i,t+1}$ *is independent of* Y_i^{t-1}, k_i^{t-1}, m_i^{t-1}, *and* X_i^t *conditional on* Y_{it}, α_i, $k_{i,t+1}$, k_{it}, $X_{i,t+1}$, *and* m_{it}.

Assumption 2 consists of two first-order Markov conditions. In part (i), log-earnings Y_{it} are allowed to affect the probability to change job directly between t and $t+1$, but the previous earnings $Y_{i,t-1}$ do not have a direct effect.[6] Similarly, in part (ii), log-earnings $Y_{i,t+1}$ may depend on the first lag of log-earnings Y_{it}, and on the current and lagged firm classes $k_{i,t+1}$ and k_{it}, but not on the further past such as $Y_{i,t-1}$ and $k_{i,t-1}$. Note that, unlike in the static model, Assumption 2(ii) restricts the evolution of log-earnings within as well as between jobs.

As a simple dynamic extension of (1), one may consider the following specification for the earnings of job movers between $t-1$ and t (i.e., $m_{i,t-1} = 1$):

$$Y_{it} = \rho_t Y_{i,t-1} + a_{1t}(k_{it}) + a_{2t}(k_{i,t-1}) + b_t(k_{it})\alpha_i + X_{it}'c_t + v_{it}, \tag{2}$$

where $\mathbb{E}(v_{it}|\alpha_i, k_i^t, m_i^{t-1}, Y_i^{t-1}, X_i^t) = 0$. Here, log-earnings after a job move may depend on earnings and firm class in the previous job.

[5]While both parts in Assumption 1 are needed to identify the full model, restrictions on dependence are not needed to identify parameters such as $a_t(k)$, $b_t(k)$, and c_t in (1).

[6]Assumption 2(i) allows $X_{i,t+1}$ to be drawn from a distribution that depends on Y_{it} as well as α_i, X_{it}, m_{it}, and $k_{i,t+1}$. Our identification arguments can be extended to this case, and estimation could allow for sequentially exogenous individual characteristics, such as job tenure.

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2.3. *Links With Theoretical Models*

In this subsection, we study whether our assumptions are compatible with various theoretical models of the labor market. We consider models that abstract from hours of work, so we refer to earnings and wages indistinctively.

Models Where the Relevant State Space Is (α, k_t)

We first consider models where wages are a function, possibly nonlinear or non-monotonic, of the worker type α, the firm class k_t, and a time-varying effect, say ε_t, where ε_t does not affect mobility decisions. This structure is compatible, for instance, with wage posting models (as in Burdett and Mortensen (1998), Delacroix and Shi (2006), or Shimer (2005)), where the wage paid to a worker does not have any history dependence and ε_t is classical measurement error or an i.i.d. match effect realized after mobility. This means that, while allowing for rich mobility and earnings patterns, such models are compatible with Assumption 1 of our static model.

Similarly, Assumption 1 is compatible with models where the wage is set as the outcome of a bargaining process between the firm and the worker under certain conditions on the worker's outside option. For example, this is the case in Shimer and Smith (2000), where the outside option is unemployment since workers always go through unemployment before finding a new job; see also Hagedorn, Law, and Manovskii (2017). In such sorting models, specifying the wage function in a way that allows for interactions between worker types and firm classes is key, since earnings may be non-monotonic in firm productivity and different workers rank identical firms differently. Our static model can accommodate both features.

Models With Markovian Match Effects and State Dependence

In dynamic models, workers often move based on the realization of the match effect ε_t, which is allowed to be serially correlated. Alternatively, ε_t may be thought of as a scalar human capital process. This is compatible with the assumptions of our dynamic model provided ε_t is first-order Markov; see Assumption 2. For example, in a wage posting model with match-specific heterogeneity, workers may observe potential wages before deciding whether or not to move. While incompatible with Assumption 1, this is perfectly consistent with the dynamic model's assumptions provided mobility, the new firm's class, and the new wage are *jointly* first-order Markov.

To see this formally, consider an agent in period t with wage Y_t and firm class k_t. She draws an offer, (Y^*_{t+1}, k^*_{t+1}), jointly with a potential wage $\widetilde{Y}_{t+1}$ she would get should she decide not to move, all of which may depend on the current wage Y_t, firm class k_t, and type α. The decision to move is based on all this information. The realized firm class is then either $k_{t+1} = k_t$ with associated wage $\widetilde{Y}_{t+1}$, or $k_{t+1} = k^*_{t+1}$ with wage Y^*_{t+1}, depending on the outcome of the mobility decision. Assumption 2 is satisfied in this model, since the effective conditioning set is (α, Y_t, k_t).

Our dynamic model encompasses other mechanisms, such as endogenous search intensity along the lines of Bagger and Lentz (2014), where the previous wage may affect offers through an endogenous search decision. It also encompasses sequential contracting as in Postel-Vinay and Robin (2002), where the Bertrand competition is captured by the fact that the outside offer Y^*_{t+1} and the firm's wage counteroffer $\widetilde{Y}_{t+1}$ may depend on each other, and (α, Y_t, k_t) are sufficient statistics for the history. Related examples are contract posting models (as in Burdett and Coles (2003), Shi (2008)), where the optimal contract is a tenure contract.

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In the setting of Assumption 2, the wage conditional on moving depends on the past wage and the past firm. Our dynamic model allows for these selection effects. However, recovering underlying primitives such as distributions of wage offers Y^*_{t+1} would require making additional assumptions. In the absence of those, our framework allows one to identify the distributions of *realized* wages for job movers and stayers, as a function of worker and firm heterogeneity.

Time Effects

Our static and dynamic models allow distributions to depend unrestrictedly on calendar time. Lise and Robin (2013) developed a model of sorting in a labor market with sequential contracting and aggregate shocks. Present values and earnings are functions of worker and firm heterogeneity, an aggregate state, and the current bargaining position. Assumption 2 of our dynamic model is satisfied in this setting.

Outside Our Framework

However, non-Markovian earnings structures will violate the assumptions of our dynamic model. This will happen if the structural model allows for permanent-transitory earnings dynamics conditional on worker types, as in Hall and Mishkin (1982), for example. This will also happen in models that combine a sequential contracting mechanism (à la Postel-Vinay and Robin (2002)) with a match-specific effect. In this case, agents need to keep track of both the match effect and the bargaining position, so the one-to-one mapping between earnings and the value to the worker no longer holds, making mobility decisions potentially dependent on the whole history of wages. Such environments are not nested in our framework, which only allows for uni-dimensional time-varying effects ε_t.

3. IDENTIFICATION

In this section, we provide conditions for identification of earnings distributions for all worker types and firm classes, and worker type distributions for all firm classes, given two periods in the static model and four periods in the dynamic model. The analysis is conditional on a partition of firms into classes. In the next section, we will show how to consistently estimate class membership $k(j)$, for each firm j.

3.1. *Intuition in an Interactive Regression Model*

We first provide an intuition for identification of complementarities in a stationary specification of the interactive regression model of equation (1) with $T = 2$ periods, where we abstract from covariates. Consider job movers between two firms of classes k and $k' \neq k$, respectively, between periods 1 and 2. Here, we study identification in a population where there is a continuum of workers moving from k to k'.[7] Log-earnings in each period are given by

$$Y_{i1} = a(k) + b(k)\alpha_i + \varepsilon_{i1}, \qquad Y_{i2} = a(k') + b(k')\alpha_i + \varepsilon_{i2}, \tag{3}$$

[7]This intuitively means that this analysis will be relevant for data sets with a sufficient number of workers moving between firm classes. Our grouping of firms into classes is motivated by the incidental parameter bias due to low mobility. We will return to this issue in the estimation section.

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